Ex NO: 10

DATE: 19/8/26

DESIGN A CUSTOMIZE SWITCH WITH NETWORK MODULES USING CISCO PACKET TRACER

AIM:

To design and configure internetworks using routers, wireless router, DHCP Server and internet cloud in Cisco packet Tracer and Verify Connectivity

PROCEDURE:

1. Open Cisco Packet Tracer and create the required network topologies.
2. Configure Router1 with
   - Fa 0/0 -> 192.168.10.1/24
   - Fa 0/1 -> 192.168.10.1/24
   - Enable interfaces using no shutdown.
3. Configure
   - PC0 -> 192.168.10.2, Gateway 192.168.10.1
   - PC1 -> 192.168.20.2, Gateway 192.168.20.1
4. Connect PCs to the router using Copper Straight-through Cables and Send a Simple PDU from PC0 to PC1
5. Create the Second Topology using PC, wireless Router, Laptop, Cable Modem, Internet cloud and Cisco.com Server.
6. Connect the devices using Straight-through And Coaxial Cables as required.

7. Configure Wireless Router:

- LAN IP: 192.168.0.1
- Mask: 255.255.255.0
- SSID: HomeNetwork
- Enable DHCP
- DNS: 208.67.220.220

8. Connect the laptop to homeNetwork using the WPC300N wireless module

9. Configure PC and Laptop to obtain IP addresses through DHCP

10. Configure Internet cloud and Cisco.com Server with the required modules DHCP and DNS Services.

11. Configure Cisco.com Server:

- IP: 208.67.220.220
- Mask: 255.255.255.0
- Gateway: 208.67.220.1

12. Verify Connectivity Using:

ipconfig /renew
ping Cisco.com

IP CONFIGURATION

DEVICE	IP ADDRESS	SUBNET MASK	GATEWAY
Router Fa0/0	192.168.10.1	255.255.255.0	—
Router Fa0/1	192.168.20.1	255.255.255.0	—
PC0	192.168.10.2	255.255.255.0	192.168.10.1
PC1	192.168.20.2	255.255.255.0	192.168.20.1
Wireless Router LAN	192.168.0.1	255.255.255.0	—
PC	DHCP (192.168.0.x)	255.255.255.0	192.168.0.1
Laptop	DHCP	255.255.255.0	192.168.0.1
Cisco.com Server	208.67.220.220	255.255.255.0	208.67.220.1

WIRELESS ROUTER & DHCP:

Wireless Router provides WiFi, SSID and internet connectivity, DHCP automatically assigns IP, gateway and DNS.

Pinging Cisco.com [208.67.220.220]

Reply from 208.67.220.220 : bytes = 32 time < 1ms TTL = 255

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Reply from 208.67.220.220 : bytes = 32 time < 1ms TTL = 255

Packet Sent = 4 , Received = 4 , Lost = 0 (0% Loss)

SIGNIFICANCE OF DHCP:

DHCP provides network settings automatically and reduces manual configuration and IP conflicts

NETWORK TOPOLOGY:

PC0 — Router — PC1

PC — WirelessRouter — cable Modem — Internet

Cloud — Server

Laptop

RESULT:

The internetworks were successfully designed and configured in Cisco Packet Tracer. Connectivity was verified using PDU and ping and DHCP successfully assigned IP configuration